Chapter 2

Examples

Generalized Pseudo code

Pseudocode is a way to describe the logic of a computer program or algorithm using plain, simple, and structured language that looks like code — but isn't written in a real programming language.

Language Element	*Generalized Pseudocode Construct*
Comment	‘ <text>
Data structure declaration	Type <type name> <list of field descriptions> End <type name>
Data declaration	Dim <variable> As <type>
Assignment statement	<variable> = <expression>
Input	Input (<variable list>)
Output	Output (<variable list>)
Condition	<expression> <relational operator> <expression>
Compound condition	<Condition> <logical connective> <Condition>
Sequence	statements in sequential order

Simple selection	If <condition> Then
	<then clause>
	EndIf

Selection	If <condition> Then <then clause> Else <else clause> EndIf
Multiple selection	Case <variable> Of Case 1: <predicate> <Case clause> ... Case *n*: <predicate> <Case clause> EndCase
Counter-controlled repetition	For <counter> = <start> To <end> <loop body> EndFor
Pretest repetition	While <condition> <loop body> EndWhile

The Triangle Problem

- Problem Statement

Simple version:

✓Problem accepts three integers a,b,c as input.

✓They are the sides of the triangle.

✓The output of the program is the type of the triangle determined by the three sides :

➢Equilateral

➢Isosceles

➢Scalene

➢Not A triangle

- Improved Version

✓The triangle program accepts three integers a,b,c as input.

✓These are taken to be the sides of a triangle.

✓The integers a , b and c must satisfy the following conditions

C1. $1 \leq a \leq 200$

C2. $1 \leq b \leq 200$

C3. $1 \leq c \leq 200$

C4. $a < b + c$

C5. $b < a + c$

C6. $c < a + b$

✓The output is the type of the triangle determined by the three sides.

✓If an input value fails any of the conditions c1,c2,c3 the program notes this with an output message like "value of b is not in the range of permitted values".

✓If values of a,b,c satisfy conditions c1,c2,c3 one of four mutually exclusive outputs is *given*:

1. If all three sides are equal, the program output is equilateral.

2. If exactly one pair of sides is equal, the program output is isosceles.

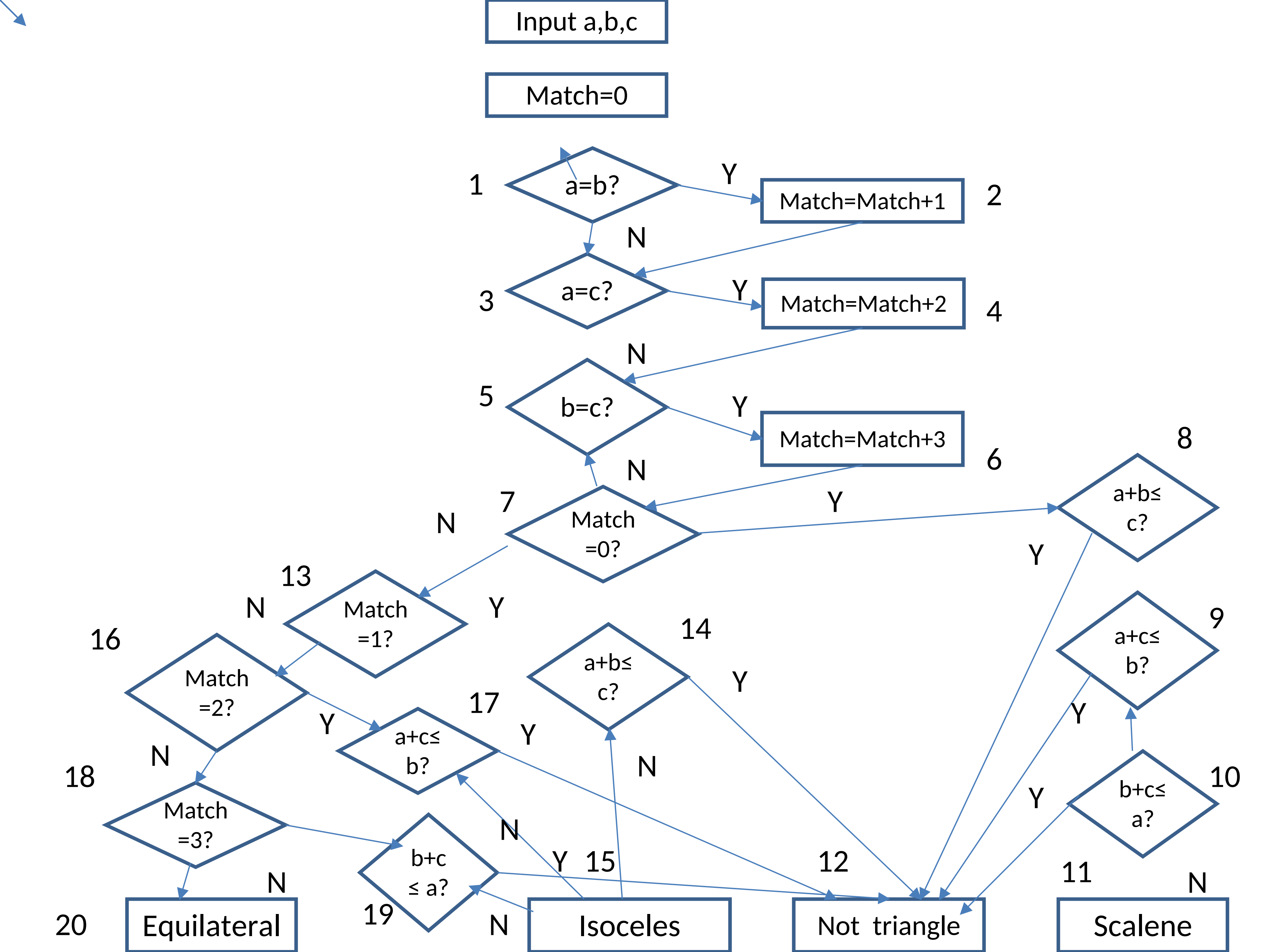
3. If no pair of sides is equal, the program output is scalene.

4. If any of conditions C4,C5,C6 is not met the program output is NotATriangle.

• The Traditional Implementation

```
Program triangle1 'Fortran-like version
`
Dim a,b,c,match  As INTEGER
`
Output("Enter the 3 integers which are sides of a triangle ")
Input(a,b,c)
Output("Side A is",a)
Output("Side B is",b)
Output("Side C is",c)
Match=0
If a=b
          Then match=match+1
EndIf
If a=c
          Then match=match+2
EndIf
If b=c
          Then match=match+3
EndIf
If match=0
                        Then  If(a+b)<=c
                               Then Output("NotATriangle")
                              Else If(b+c)<=a
                               Then Output("NotATriangle")
                              Else If(a+c)<=b
                               Then Output("NotATriangle")
                              Else Output("Scalene")
```

```
            EndIf
        EndIf
     EndIf
Else If match=1
     Then If(a+b)<=c
               Then Output("NotATriangle")
           Else Output("Isoceles")
     EndIf
Else If match=2
     Then If (a+c)<=b
           Then Output ("NotATriangle")
      Else Output("Isoceles")
EndIf
Else If match=3
     Then If (b+c)<=a
           Then Output("NotATriangle")
     Else Output("Isoceles")
EndIf
Else Output ("equilateral")
   EndIf
  EndIf
 EndIf
EndIf
`
End Triangle 1
```

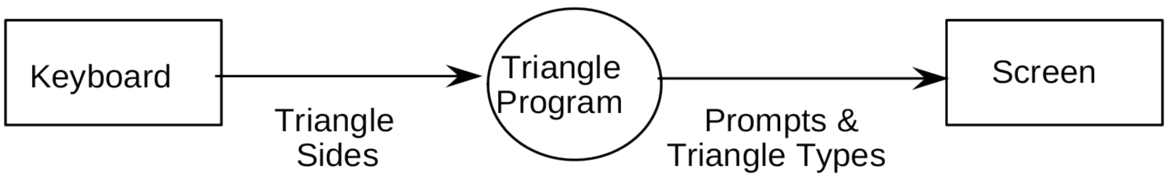


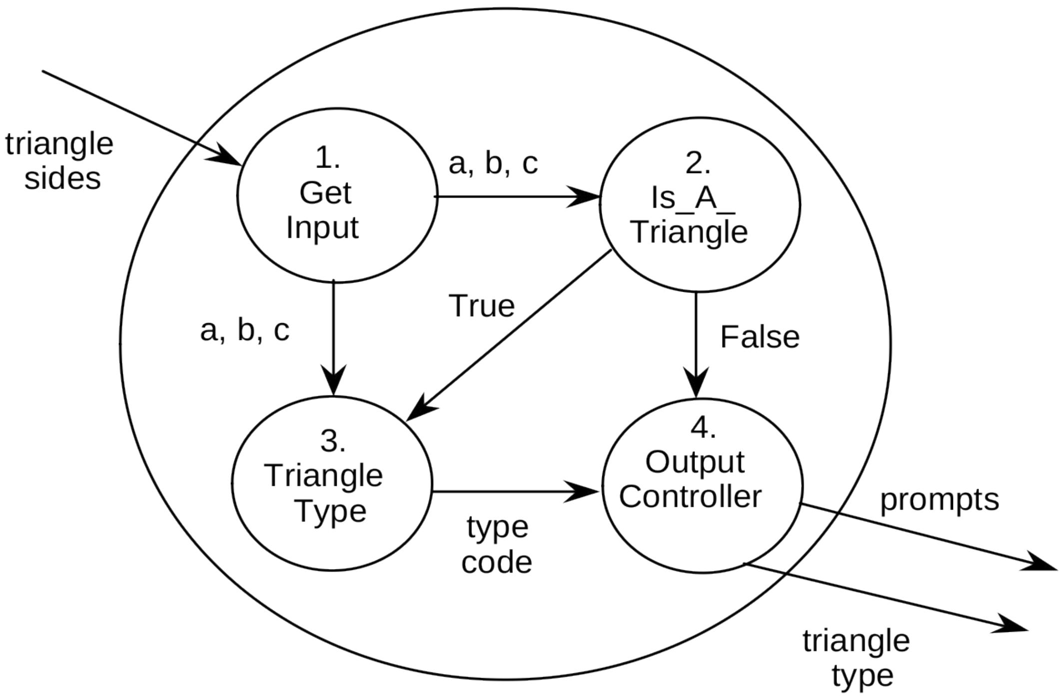
Input a,b,c
Match=0
1
a=b?
Y
Match=Match+1
2
N
3
a=c?
Y
Match=Match+2
4
N
5
b=c?
Y
Match=Match+3
6
N
7
Match
=0?
Y
8
a+b≤
c?
N
13
Match
=1?
Y
14
a+b≤
c?
Y
9
a+c≤
b?
16
Match
=2?
Y
17
a+c≤
b?
Y
N
10
b+c≤
a?
18
Match
=3?
N
19
b+c
≤ a?
15
12
11
N
20
Equilateral
Isoceles
Not triangle
Scalene

Structured Implementation

Dataflow diagram for a triangle program implementation

Triangle Program Specification





Data Dictionary

Triangle Sides : a + b + c

a, b, c are non-negative integers

type code : equilateral | isosceles | scalene | not a triangle

prompts : 'Enter three integers which are sides of a triangle'

triangle type : 'Equilateral' | 'Isosceles' | 'Scalene' | 'Not a Triangle'

```
Program triangle2 `Structured programming version of simpler specification
`
Dim a,b,c As Integer
Dim IsATriangle As Boolean
`
`Step1: Get Input
Output("Enter the 3 integers which are sides of a triangle ")
Input(a,b,c)
Output("Side A is",a)
Output("Side B is",b)
Output("Side C is",c)
`
`Step2:Is A Triangle?
If(a<b+c) AND (b<a+c) AND (c<a+b)
Then IsATriangle=True
Else IsATriangle=False
EndIf
`
`Step3:Determine Triangle Type
If IsATriangle
Then If(a=b) AND (b=c)
Then Output ("Equilateral")
Else If( a!=b) AND (a!=c) AND (b!=c)
Then Output ("Scalene")
Else Output (" Isoceles")
EndIf
EndIf
Else Output("Not a Triangle")
EndIf
`
Endtriangle2
```

```
Program triangle3 `Structured programming version of improved specification
`
Dim a,b,c As Integer
Dim c1,c2,c3,IsATriangle As Boolean
`
`Step1: Get Input
Do
Output("Enter 3 integers which are the sides of the triangle")
Input(a,b,c)
C1=(1<=a) AND (a<=200)
C2=(1<=b) AND (b<=200)
C3=(1<=c) AND (c<=200)
If NOT (c1)
Then Output("Value of a is  not in the range of permitted values")
EndIf
If NOT(c2)
Then Output("Value of b is  not in the range of permitted values")
EndIf
If NOT(c3)
Then Output("Value of c is  not in the range of permitted values")
EndIf
Until c1 AND c2 AND c3
Output("Side  A is ",a)
Output("Side B is",b)
Output("Side C is",c)
`
```

```
`Step 2: Is A Triangle?
If(a<(b+c)) AND (b<(a+c)) AND (c<(a+b))
Then IsATriangle=True
Else IsATriangle=False
EndIf
`
`Step3:Determine Triangle Type
If IsATriangle
Then If(a=b) AND(b=c)
Then Output("Equilateral")
Else If (a!=b) AND (a!=c) AND (b!=c)
Then Output ("Scalene")
Else Output ("Isoceles")
EndIf
EndIf
Else Output ("Not a Triangle")
EndIf
`
End triangle3
```

•

The NextDate Function

- NextDate is a function of three variables: month, day and year.
- It returns the date of the day after the input date.
- The month,day and year variables have integer values subject to these conditions:

C1 $1 \le month \le 12$
C2 $1 \le day \le 31$
C3 $1812 \le year \le 2012$

```
Program NextDate
Dim tomorrowDay,tomorrowMonth,tomorrowYear As Integer
Dim day,month,year As Integer
Output ("Enter today's date in the form MM DD YYYY")
Input (month,day,year)
Case month Of
Case 1: month Is 1,3,5,7,8,Or 10: '31 day months (except Dec.)
    If day < 31
        Then tomorrowDay = day + 1
        Else
            tomorrowDay = 1
            tomorrowMonth = month + 1
    EndIf
Case 2: month Is 4,6,9,Or 11 '30 day months
    If day < 30
        Then tomorrowDay = day + 1
        Else
            tomorrowDay = 1
            tomorrowMonth = month + 1
    EndIf
Case 3: month Is 12: 'December
    If day < 31
        Then tomorrowDay = day + 1
        Else
            tomorrowDay = 1
            tomorrowMonth = 1
            If year = 2012
                Then Output ("2012 is over")
                Else tomorrow.year = year + 1
            EndIf
    EndIf
```

```
Case 4: month is 2: 'February
    If day < 28
        Then tomorrowDay = day + 1
        Else
            If day = 28
                Then
                    If ((year MOD 4)=0)AND((year MOD 400)š0)
                        Then tomorrowDay = 29 'leap year
                        Else            'not a leap year
                            tomorrowDay = 1
                            tomorrowMonth = 3
                    EndIf
                Else   If day = 29
                            Then tomorrowDay = 1
                                tomorrowMonth = 3
                            Else   Output("Cannot have Feb.", day)
                        EndIf
            EndIf
    EndIf
EndCase
Output ("Tomorrow's date is", tomorrowMonth,
                tomorrowDay, tomorrowYear)
End NextDate
```

```
Program NextDate2      Improved version
`
Dim tomorrowDay,tomorrowMonth, tomorrowYear As Integer
Dim day,month,year As Integer
Dim c1, c2, c3 As Boolean
`
Do
   Output ("Enter today's date in the form MM DD YYYY")
   Input (month,day,year)
   c1 = (1 <= day) AND (day <= 31)
   c2 = (1 <= month) AND (month <= 12)
   c3 = (1812 <= year) AND (year <= 2012)
   If NOT(c1)
     Then     Output ("Value of day not in the range 1..31")
   EndIf
   If NOT (c2)
   EndIf
     Then     Output ("Value of month not in the range 1..12")
   If NOT(c3)
     Then  Output ("Value of year not in the range 1812..2012")
   EndIf
Until c1 AND c2 AND c3
Case month Of
Case 1: month Is 1,3,5,7,8, Or 10: '31 day months (except Dec.)
   If day < 31
     Then tomorrowDay = day + 1
     Else
       tomorrowDay = 1
       tomorrowMonth = month + 1
   EndIf
```

```
Case 2: month Is 4,6,9, Or 11 '30 day months
  If day < 30
    Then tomorrowDay = day + 1
    Else
       If day =30
         Then  tomorrowDay = 1
                 tomorrowMonth = month + 1
         Else  Output ("Invalid Input Date")
       EndIf
  EndIf
Case 3: month Is 12: 'December
  If day < 31
    Then tomorrowDay = day + 1
    Else
       tomorrowDay = 1
       tomorrowMonth = 1
       If year = 2012
         Then Output ("Invalid Input Date")
         Else tomorrow.year = year + 1
       EndIf
  EndIf
```

```
Case 4: month is 2: 'February
  If day < 28
    Then tomorrowDay = day + 1
    Else
      If day = 28
        Then
          If (year is a leap year)
             Then tomorrowDay = 29 'leap day
             Else  'not a leap year
                 tomorrowDay = 1
                    tomorrowMonth = 3
             EndIf
          Else
             If day = 29
                Then
                   If (Year is a leap year)
                      Then tomorrowDay = 1
                          tomorrowMonth = 3
                Else
                   If day > 29
                      Then Output ("Invalid Input Date")
                   EndIf
                EndIf
             EndIf
         EndIf
  EndIf
EndCase
Output ("Tomorrow's date is", tomorrowMonth, tomorrowDay,
tomorrowYear)

End NextDate2
```

The commission problem

- A rifle salesperson sold rifle locks,stocks and barrels made by a gun-smith.
- Locks cost $45,stocks cost $30,and barrels cost $25.
- The salesperson could sell a maximum of 70 locks,80 stocks and 90 barrels in a month.
- After each town visit the salesperson sent a telegram with the number of locks,stocks and barrels sold in the town.
- At the end of a month, the salesperson sent a short telegram showing -1 lock sold.
- The gun-smith then knew the sales for the month were complete.

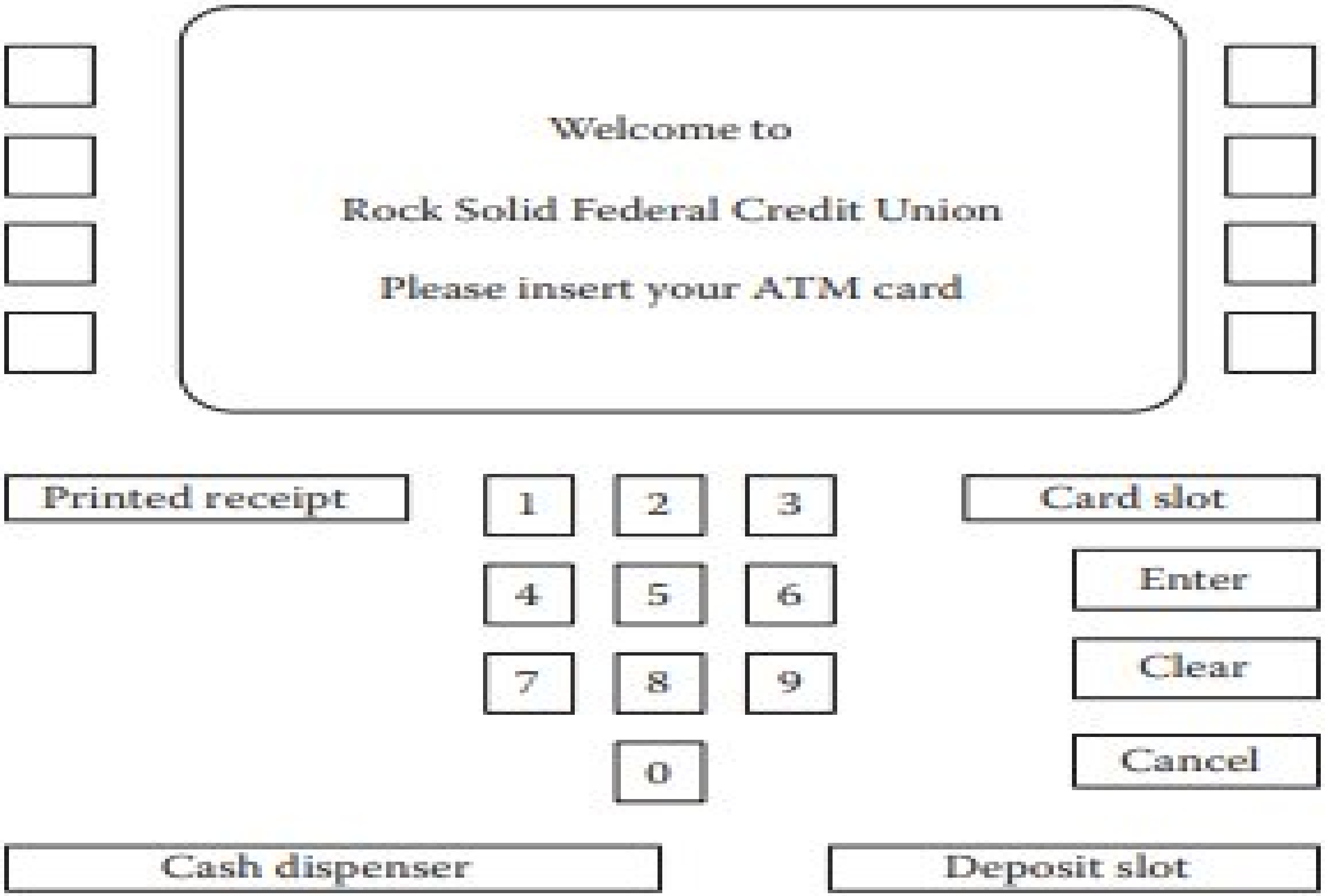
- The commission was then computed as follows:
  - ✓10% on sales up to $1000
  - ✓15% on the next $800
  - ✓20% on any sales in excess of $1800.

- The commission program produced a monthly sales report that gave the total number of locks,stocks and barrels sold,the salesperson's total dollar sales, and finally the commission.

```
1.Program Commission (INPUT,OUTPUT)
`
2.Dim locks,stocks,barrels  As Integer
3.Dim lockPrice,stockPrice,barrelPrice As Real
4.Dim totalLocks,totalStocks,totalBarrels As Integer
5.Dim lockSales,stockSales,barrelSales As Real
6.Dim sales,commission:REAL
`
7.lockPrice=45.0
8.stockPrice=30.0
9.barrelPrice=25.0
10.totalLocks=0
11.totalStocks=0
12.totalBarrels=0
`
13.Input(locks)
14.While NOT(locks=-1) `Input device uses -1 to indicate end of data
15.Input (stocks,barrels)
16.totalLocks=totalLocks+locks
17.totalStocks=totalStocks+stocks
18.totalBarrels=totalBarrels+barrels
19.Input(locks)
20.EndWhile
```

```
`
21.Output("Locks sold",totalLocks)
22.Output("Stocks sold",totalStocks)
23.Output("Barrels sold",totalBarrels)
`
24.lockSales=lockPrice*totalLocks
25.stockSales=stockPrice*totalStocks
26.barrelSales=barrelPrice*totalBarrels
27.sales=lockSales+stockSales+barrelSales
28.Output("Total sales",sales)
`
29.If (sales > 1800.0)
30.Then
31.Commission=0.10*1000.0
32.Commission=commission+0.15*800.0
33.Commission=commission+0.20*(sales-1800.0)
34.Else If(sales > 1000.0)
35.Then
36.Commission=0.10*1000.0
37.Commission=commission+0.15*(sales-1000.0)
38.Else
39.commission=0.10*sales
40.EndIf
41.EndIf
42.Output ("Commission is $",commission)
`
43.End Commission
```

The SATM System



Screen 1

Welcome
please insert your
ATM card

Screen 2

Please enter your PIN
____

Screen 3

Your PIN is incorrect.
Please try again.

Screen 4

Invalid ATM card. It will
be retained.

Screen 5

Select transaction:
balance >
deposit >
withdrawal >

Screen 6

Balance is
$dddd.dd

Screen 7

Enter amount.
Withdrawals must
be multiples of $10

Screen 8

Insufficient Funds!
Please enter a new
amount

Screen 9

Machine can only
dispense $10 notes

Screen 10

Temporarily unable to
process withdrawals.
Another transaction?

Screen 11

Your balance is being
updated. Please take
cash from dispenser.

Screen 12

Temporarily unable to
process deposits.
Another transaction?

Screen 13

Please insert deposit
into deposit slot.

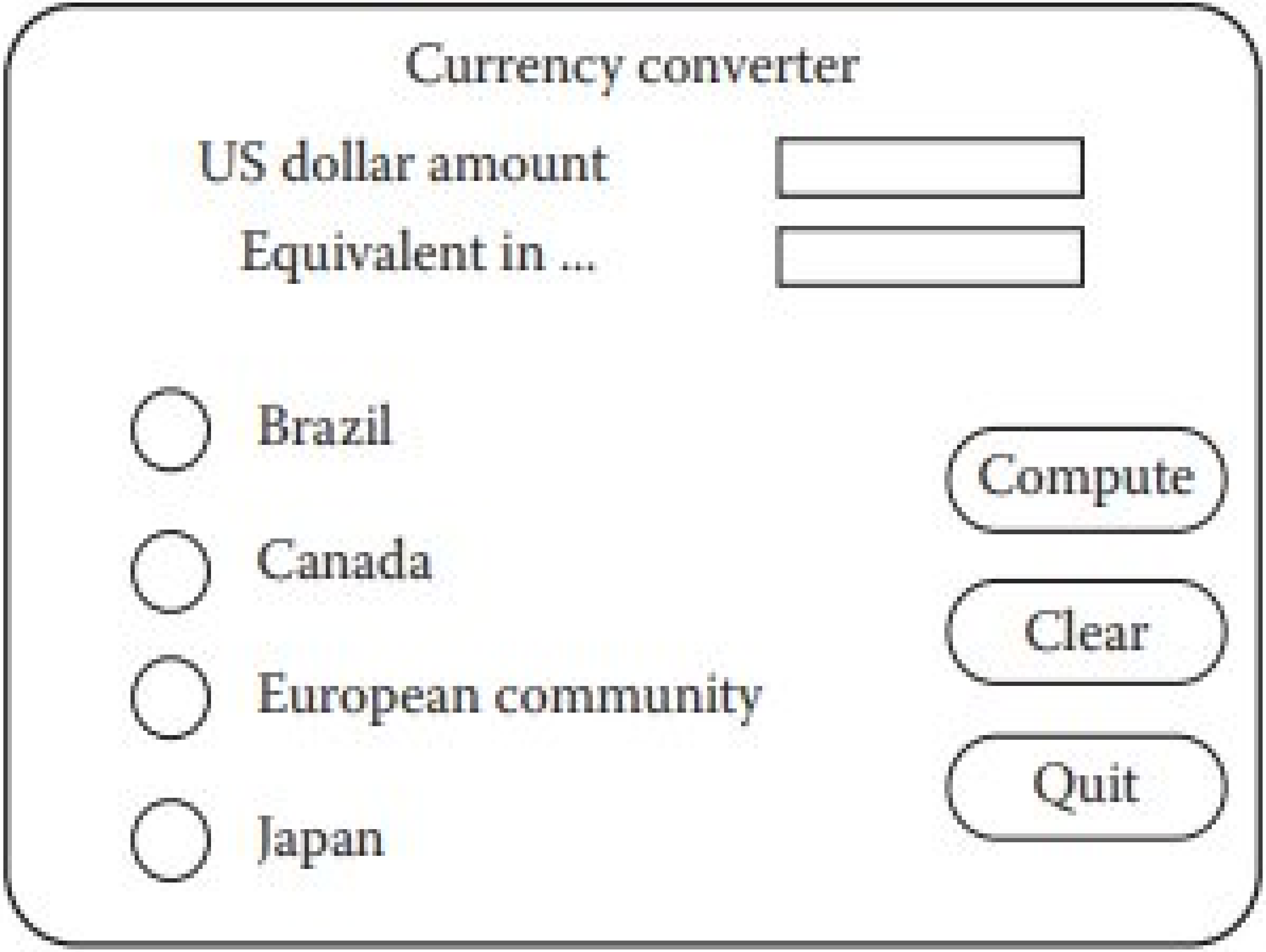
Screen 14

Your new balance is
being printed. Another
transaction?

Screen 15

Please take your
receipt and ATM card.
Thank you.

The currency converter



Saturn Windshield Wiper Controller

C1	Lever	OFF	INT	INT	INT	LOW	HIGH
C2	Dial	n/a	1	2	3	n/a	n/a
A1	Wiper	0	4	6	12	30	60